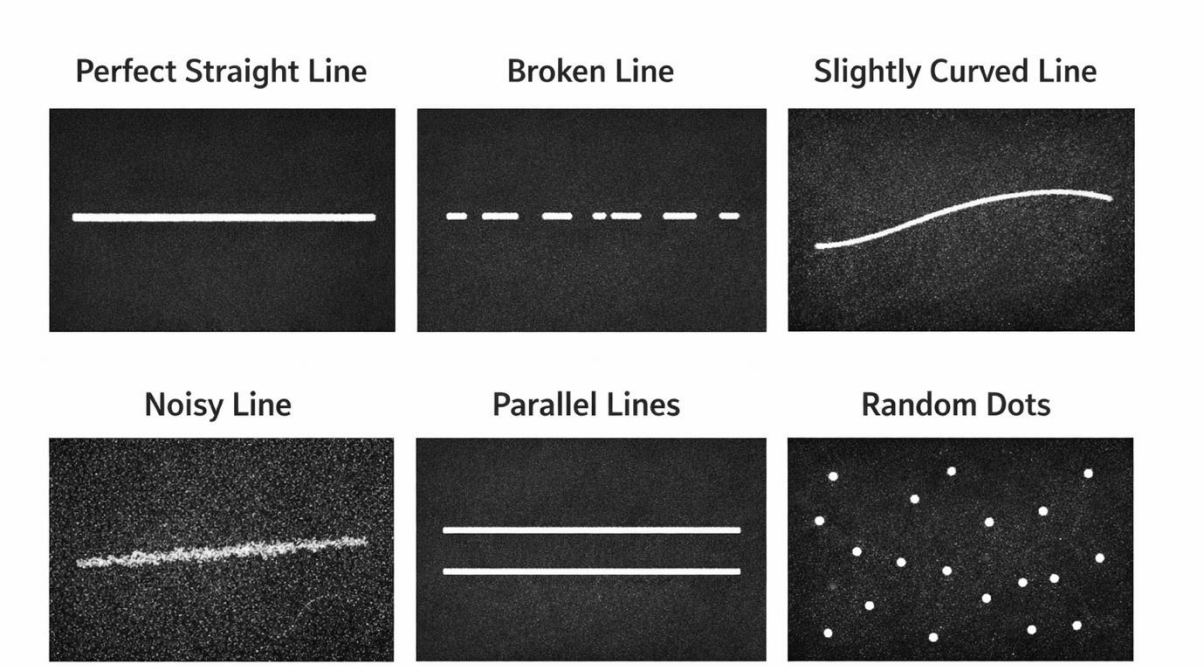
Unit 3 - Session 1

SLO 1: Understand the principles of line detection in images.

Duration: 10 Minutes Activity

Question:

1. Consider **6 printed or projected image patches / sketches**:



2. For **each image**, ask students to answer **only one question**:

"Can a computer detect a straight line here?"

3. Students respond by:

- Raising **YES** / **NO** cards
- OR showing thumbs 👍 / 👎

4. After each vote, briefly discuss the decision.

1. Perfect Straight Line – YES: The line is clear, continuous, and straight, so a computer can easily detect it.

2. Broken Line – NO: The line is not continuous because there are gaps, making straight-line detection more difficult.
3. Slightly Curved Line – NO: The line is curved rather than perfectly straight, so it may not be detected as a straight line.
4. Noisy Line – YES: The main line is still visible despite the noise, so a computer can detect the underlying straight line.
5. Parallel Lines – YES: Both lines are clear and straight, so a computer can detect them as separate straight lines.
6. Random Dots – NO: There is no continuous straight-line pattern; the dots are randomly placed.